

Noncaloric monosaccharides induce excessive sprouting angiogenesis in zebrafish via foxo1a-marcksl1a signal

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Abstract

Artificially sweetened beverages containing noncaloric monosaccharides were suggested as healthier alternatives to sugar-sweetened beverages. Nevertheless, the potential detrimental effects of these noncaloric monosaccharides on blood vessel function remain inadequately understood. Presently, we have established a zebrafish model that exhibits significant excessive angiogenesis induced by high glucose, resembling the hyperangiogenic characteristics observed in proliferative diabetic retinopathy (PDR). Utilizing this model, we observed that glucose and noncaloric monosaccharides could induce excessive formation of blood vessels, especially intersegmental vessels (ISVs). The excessively branched vessels were observed to be formed by ectopic activation of quiescent endothelial cells (ECs) into tip cells. Single-cell transcriptomic sequencing analysis of the endothelial cells in the embryos exposed to high glucose revealed an augmented ratio of capillary ECs, proliferating ECs, and a series of upregulated proangiogenic genes. Further analysis and experiments validated that foxo1a mediated the excessive angiogenesis induced by monosaccharides by down-regulating the expression of marcksl1a. This study has provided new evidence showing the negative effects of noncaloric monosaccharides on the vascular system and the underlying mechanisms.

eLife assessment

The manuscript addresses an **important** subject regarding the effects of the consumption of artificially sweetened beverages. The authors used zebrafish embryos treated with high concentrations of monosaccharides and observed hyperangiogenic characteristics similar to those seen in proliferative diabetic retinopathy. Their findings suggest that caloric and noncaloric monosaccharides induce excessive angiogenesis via the foxo1a-Marcksl1a pathway, but the evidence is currently **incomplete**.

Introduction

Diabetes mellitus (DM) encompasses a group of chronic diseases characterized by elevated blood glucose levels. Among patients with DM, cardiovascular complications, especially the direct and indirect effects of hyperglycemia on the human vascular network, persist as the primary cause of morbidity and mortality [1]. The harmful effects of hyperglycemia are closely associated with both microvascular and macrovascular complications, including retinopathy, nephropathy, neuropathy, atherosclerosis, ischemic heart disease, stroke, and peripheral artery disease [2, 3]. Endothelial dysfunction (ECD) is a systemic pathological state exhibiting disrupted integrity, adhesion, altered proliferation capacity, migration, tube formation, and more [4, 5]. High blood glucose levels over long periods have been demonstrated to be associated with vascular dysfunction both *in vivo* and *in vitro* [6, 7].

The epidemiological evidence has indicated the positive correlation between risks of cardiovascular disease and DM with the consumption of sugar-sweetened beverages (SSBs) and 100% fruit juices, thereby emphasizing the concerns for the adverse effects of sugar intake on cardiometabolic risk factors, regardless of whether the sugar is added or naturally occurring [8-12]. Artificially sweetened beverages (ASBs), which incorporate noncaloric sweeteners or low-caloric additives, have been suggested as healthy alternatives to SSBs [13]. ASBs contain sugar alcohols and polyols, such as sorbitol, xylitol, maltitol, mannitol, erythritol, isomalt, and lactitol. The consumption of ASBs worldwide has gradually increased in recent years [13-15]. However, accumulating studies in the last decade suggested that ASB consumption might be associated with an increased risk of cardiovascular events and diabetes [12, 16-23]. Nevertheless, the underlying mechanisms responsible for these findings remain insufficiently documented.

The zebrafish has been recognized as a valuable animal model for studying metabolic diseases, such as hyperglycemia and diabetic complications, due to its functional conservation in glycol metabolism, pancreas structure, glucose homeostasis, adipose biology, and genetic similarities to mammals [24-26]. The combination of embryonic transparency and transgenic lines, wherein endothelial cells are labeled specifically with fluorescent proteins, facilitates the high-resolution imaging analysis of vascular formation *in vivo*. Immersion of zebrafish in glucose solution has been found to induce diabetic complications, including vascular dysfunction [27-30]. Several recent studies have investigated the effects of high glucose on vascular function in the zebrafish model [30-32]. However, the association between noncaloric monosaccharides and vascular dysfunctions, such as excessive angiogenesis, has not been elucidated. Here, we have successfully established a short-term zebrafish model that exhibits significantly excessive angiogenesis similar to the phenotypes observed in proliferative diabetic retinopathy (PDR) induced by glucose treatment. Using this model, we examined the effects of noncaloric monosaccharides on blood vessel development and investigated the molecular mechanisms. Our results provided new evidence for the negative roles of caloric and noncaloric monosaccharides on vascular development.

Results

Establishment of a short-term model of high-glucose-induced excessive angiogenesis

To establish the short-term zebrafish hyperangiogenic model induced by high-glucose treatment, we immersed the *Tg(fli1aEP:EGFP-CAAX)^{ntu666}* embryos, a transgenic line wherein the endothelial cells were labeled with membrane-bound GFP (Supplementary Figure 1), in glucose solution within a wide range of concentrations and time windows (Supplementary Figure 2). We

subsequently measured the glucose concentration in the embryos and found that the glucose concentration in the embryos treated with high glucose was significantly higher than that in the control group (Supplementary Figure 3). We observed that exposing zebrafish embryos at either 24 hours post fertilization (hpf) or 48 hpf to a 6% D-glucose treatment for a duration exceeding 48 hours led to dramatically increased formation of blood vessels (**Figure 1**[↗](#), Movie 1 and 2), especially intersegmental vessels (ISVs) in the indicated area (**Figure 1b**[↗](#)). The hyperbranched endothelial cells were observed to sprout from existing vessels, including the ISVs, dorsal aorta (DA), and dorsal lateral anastomotic vessel (DLAV) (**Figure 1**[↗](#)) in embryos treated with high glucose. Additionally, these ectopically branched angiogenic sprouts were not perfused by blood flow. Despite the abnormal vessel formation, no significant developmental defects were observed in these treated embryos when examined under a bright field microscope (Supplementary Figure 4a-c). Moreover, no excessive angiogenic phenotype was observed in the embryos treated with 1%, 2%, 3%, and 4% D-glucose within the corresponding time frame (Supplementary Figure 5). In addition, 6% sucrose treatment at the same condition did not lead to any noticeable abnormal phenotype of vessels (Supplementary Figure 4d-h).

Fructose and noncaloric monosaccharides induce excessive angiogenesis

Fructose is a ketonic monosaccharide that is an energy source for living organisms. Therefore, our study investigated the potential effects of fructose on vascular dysfunction in comparison to glucose. The result demonstrated fructose-induced excessive angiogenesis in zebrafish embryos (Supplementary Figure 6). Wondering whether the effects of glucose and fructose on vascular development were mediated by metabolic events, we then conducted the same tests by using other noncaloric monosaccharides, including L-glucose, D-mannose, D-ribose, and L-arabinose, which could not be digested by animals. Interestingly, we observed that all these noncaloric monosaccharides could induce excessive angiogenesis, among which the L-glucose purchased from two companies resulted in a similar phenotype as efficiently as D-glucose did (**Figure 2a-h**[↗](#)). However, the disaccharides, including lactose, maltose, and sucrose, which were also tested, did not cause significant excessive angiogenic phenotype (Supplementary Figure 4e, f; Supplementary Figure 7). In addition, we also tested the effects of pyruvic acid but did not observe the excessive angiogenic phenotype in the embryos treated with pyruvic acid solution at 50 nm~50 μ m concentration (Supplementary Figure 8). Furthermore, we examined the arterial and venous identity of the hyperbranched vessels via live imaging analysis of the high glucose-treated *Tg(flt1:YFP::kdrl:ras-mCherry)* line, in which the YFP expression in the artery was relatively higher than that in the vein [[33](#)[↗](#)]. The result revealed that the hyperbranched ectopic vessels comprised arteries and veins (**Figure 2i, j**[↗](#)).

High glucose promotes quiescent endothelial differentiation into tip cells

Given that a high-glucose shock has been observed to induce excessive angiogenesis in 48 hpf embryos, it was hypothesized that the shock might play a crucial role in regulating the differentiation of quiescent endothelial cells (ECs) into active tip cell-like cells and their subsequent behaviors. To investigate whether this was the case, we observed the behaviors of these ECs by confocal time-lapse imaging analysis. As shown in the result, in the control *Tg(fli1aEP:EGFP-CAAX)^{ntu666}* embryos, no significant activation of tip cells in the angiogenic sprouts was observed in the generated ISVs, DA, and DLAV in the embryos aged from 48 hpf to 5 dpf. Moreover, only a few ECs in established ISVs, DA, and DLAV extended filopodia, which quickly retracted (**Figure 3a-c**[↗](#), Movie 3). However, many ECs initiated sprouting angiogenesis in the high glucose-treated embryos, extended dynamic filopodia to sense the surroundings, and formed excessive ectopic blood vessels (**Figure 3d-e**[↗](#), Movie 4). In a snapshot, we observed that some of the ECs protruded long and intricate sprouts simultaneously (**Figure 3f**[↗](#)), and nearly all the ECs

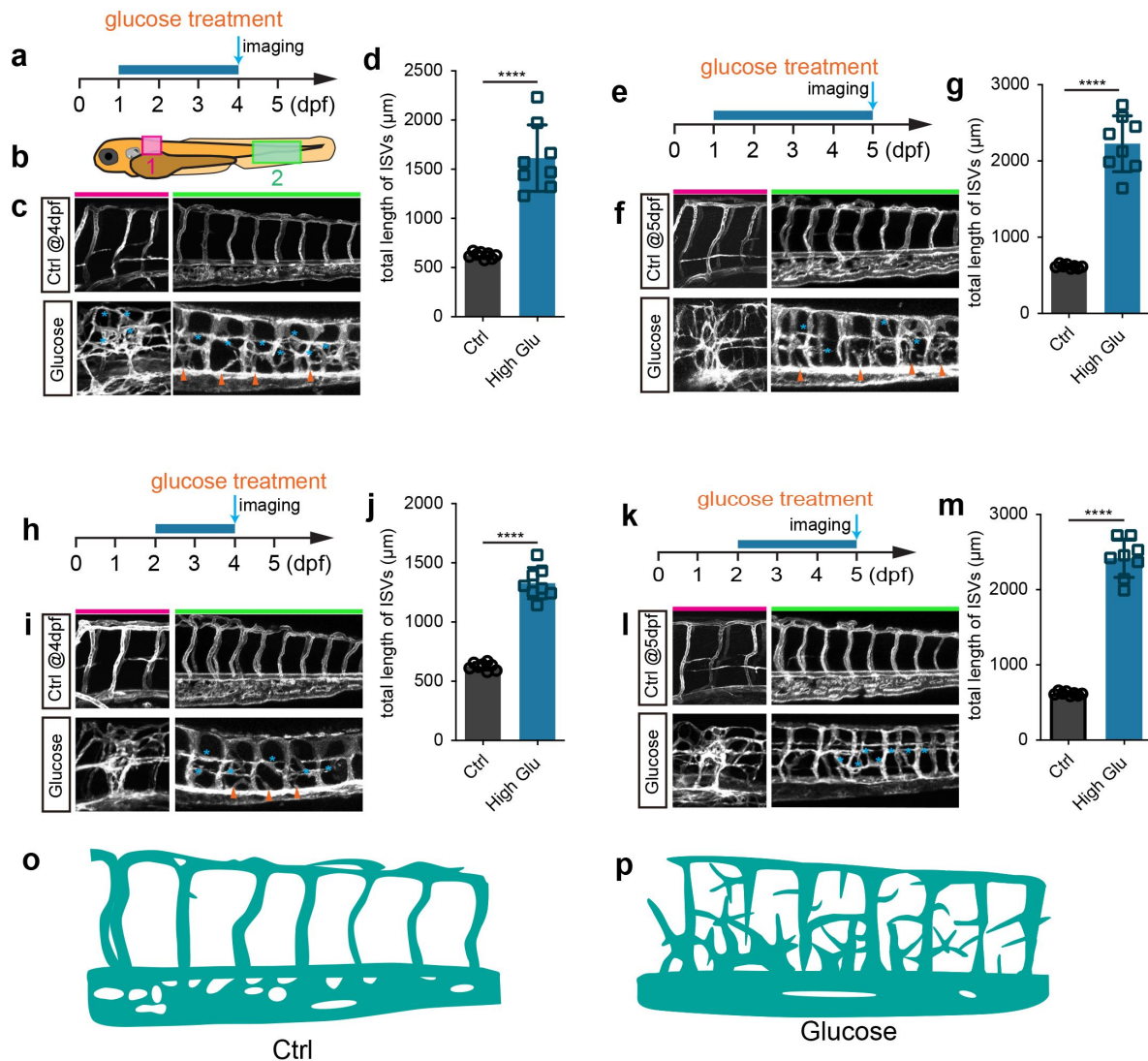


Figure 1

Glucose treatment caused excessive angiogenesis in zebrafish.

a, A diagram showing the glucose treatment time window and imaging time point. **b**, A diagram indicating the imaging positions of the zebrafish embryos. **c**, Confocal imaging analysis of the control and glucose-treated embryos. The red bar indicates position 1; the green bar indicates position 2. Arrowheads indicate the ectopic branching from the dorsal aorta. Stars indicate the ectopic vessels from ISVs and DLAVs. **d**, Statistical analysis of the total length of ISVs in control and glucose-treated embryos. t-test, **** $p < 0.0001$. **e**, A diagram showing the glucose treatment time window and imaging time point. **f**, Confocal imaging analysis of the control and glucose-treated embryos. The red bar indicates position 1; the green bar indicates position 2. Arrowheads indicate the ectopic branching from the dorsal aorta. Stars indicate the ectopic vessels from ISVs and DLAVs. **g**, Statistical analysis of the total length of ISVs in control and glucose-treated embryos. t-test, **** $p < 0.0001$. **h**, A diagram showing the glucose treatment time window and imaging time point. **i**, Confocal imaging analysis of the control and glucose-treated embryos. The red bar indicates position 1; the green bar indicates position 2. Arrowheads indicate the ectopic branching from the dorsal aorta. Stars indicate the ectopic vessels from ISVs and DLAVs. **j**, Statistical analysis of the total length of ISVs in control and glucose-treated embryos. t-test, **** $p < 0.0001$. **k**, A diagram showing the glucose treatment time window and imaging time point. **l**, Confocal imaging analysis of the control and glucose-treated embryos. The red bar indicates position 1; the green bar indicates position 2. Arrowheads indicate the ectopic branching from the dorsal aorta. Stars indicate the ectopic vessels from ISVs and DLAVs. **m**, Statistical analysis of the total length of ISVs in control and glucose-treated embryos. t-test, **** $p < 0.0001$. **o**, A diagram showing the blood vessels in position 2 indicated in panel b of control embryos. **p**, A diagram showing the blood vessels in position 2 indicated in panel b of high glucose-treated embryos.

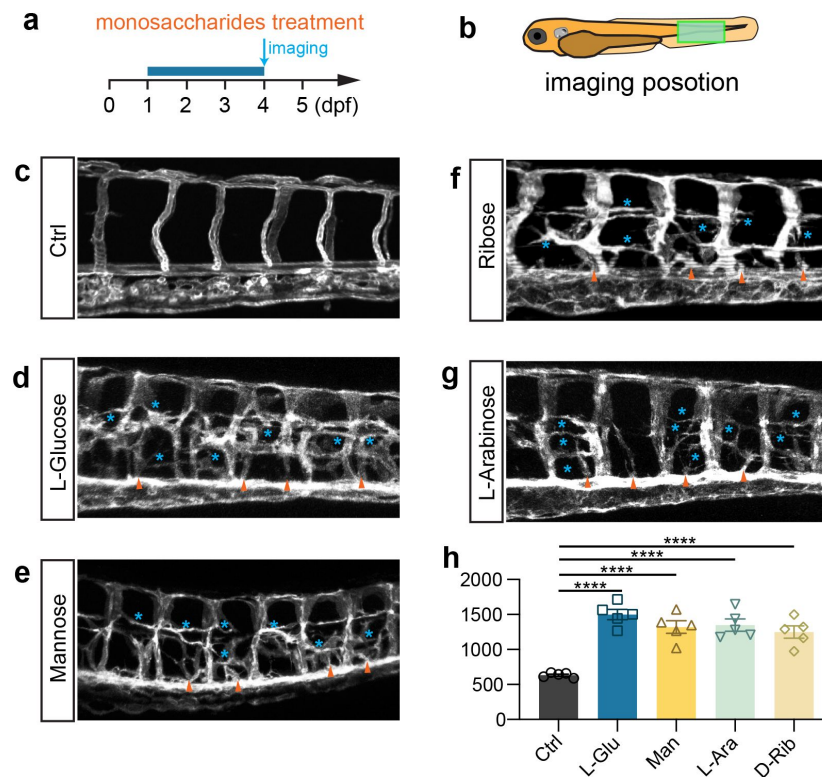


Figure 2

L-glucose and mannose treatment caused excessive angiogenesis as well.

a, A diagram showing the monosaccharides treatment time window and imaging time point. **b**, A diagram indicating the imaging position of the zebrafish embryos. **c-g**, Confocal imaging analysis of the control and monosaccharides, including L-glucose, D-mannose, D-ribose, and L-arabinose, treated embryos. Arrowheads indicate the ectopic branching from the dorsal aorta. Stars indicate the ectopic vessels from ISVs. **h**, Statistical analysis of the total length of ISVs in control and monosaccharides treated embryos. t-test, ****p<0.0001.

within an ISV underwent the outgrowth of filopodia in some extreme cases (**Figure 3g**), suggesting that the high glucose treatment induced the endothelial differentiation into tip cell-like cells. Furthermore, we observed that these outgrowths of the ectopic angiogenic sprouts could establish a connection to the neighboring sprouts and vessels and thereby form complicated vascular structures (**Figure 1c, f, i, l, p**).

Single-cell transcriptomic sequencing analysis of the endothelial cells isolated from glucose-treated embryos

We did a single-cell transcriptomic sequencing analysis to gain more insight into the potential mechanism through which glucose activates the endothelial cells. Due to the limited presence of endothelial cells within the zebrafish embryos, the analysis of these cells poses a challenge. Firstly, we isolated the EGFP-positive cells from control and high glucose-treated embryos. Following the proteolytic dissociation of embryos, the EGFP-positive cells were isolated by fluorescence-activated cell sorting (FACS). Around 300-500 zebrafish embryos were used for the ECs collection for each stage. The isolated ECs were analyzed using a large-scale scRNA-seq (10X Genomics) platform, and the pipeline is illustrated in the diagram (**Figure 4a**). Multiple criteria were applied to select the single cells, including the retention of the genes that were expressed (Unique Molecular Identifiers or UMI larger than 0) in at least 3 individual cells, the selection of cells with the gene expression count falling within the range of 500 to 3000, and the imposition of a threshold wherein the proportion of sequencing reads derived from the mitochondrial genome was limited to less than 5% (Supplementary Figure 9). Ultimately, 6051 endothelial cells were selected for further analysis (Supplementary Figure 10, Supplementary Table 1).

Through clustering analysis of gene expression, these ECs were categorized into 6 clusters using UMAP. These clusters include cluster 0, which consists of arterial and capillary ECs; cluster1, comprising endocardium; cluster2, consisting of venous and lymphatic ECs; cluster3, comprising arch ECs; cluster4, encompassing proliferating ECs; and cluster5, consisting of vesicle enriched ECs (**Figure 4b**). The endothelial marker gene *cdh5* was expressed in all the clusters (**Figure 4c**). The notch ligand *dlc* was highly expressed in arterial, capillary ECs, and arch ECs (**Figure 4d**). The *dab2* and *prox1* were mainly enriched in venous and lymphatic ECs (**Figure 4e-f**). The *cdk1*, which is a key player in cell cycle regulation, was specifically expressed in proliferating ECs (**Figure 4h**). It was revealed that the ratio of arterial and capillary ECs and proliferating ECs was increased in the high glucose-treated embryos (**Figure 4i, j**), consistent with the observation that glucose treatment resulted in excessive sprouting angiogenesis of ISVs.

Foxo1a was significantly downregulated in arterial and capillary ECs

To identify the potential molecules responsible for increasing the proportion of arterial and capillary ECs in the embryos treated with glucose, we analyzed and compared the differentially expressed genes (DEGs) in arterial and capillary ECs of control and glucose-treated ECs. The results revealed that 1201 genes were up-regulated and 523 genes were down-regulated significantly (**Figure 5a**). GO analysis revealed that these DEGs were enriched in several biological processes, including regulation of actin filament organization, blood vessel morphogenesis, development, angiogenesis, etc. (**Figure 5b**).

Subsequently, we searched for transcription factors among the genes involved in the aforementioned biological processes that might participate in inducing excessive angiogenesis. It has been reported that the loss of function of *foxo1a* led to excessive angiogenesis [34, 35]. Our study also revealed that *foxo1a* was significantly downregulated in arterial and capillary ECs after high glucose treatment compared to the ECs marker gene *pecam1* (**Figure 5c-e**). The *in situ* hybridization (ISH) experiments further confirmed the decrease in *foxo1a* expression following

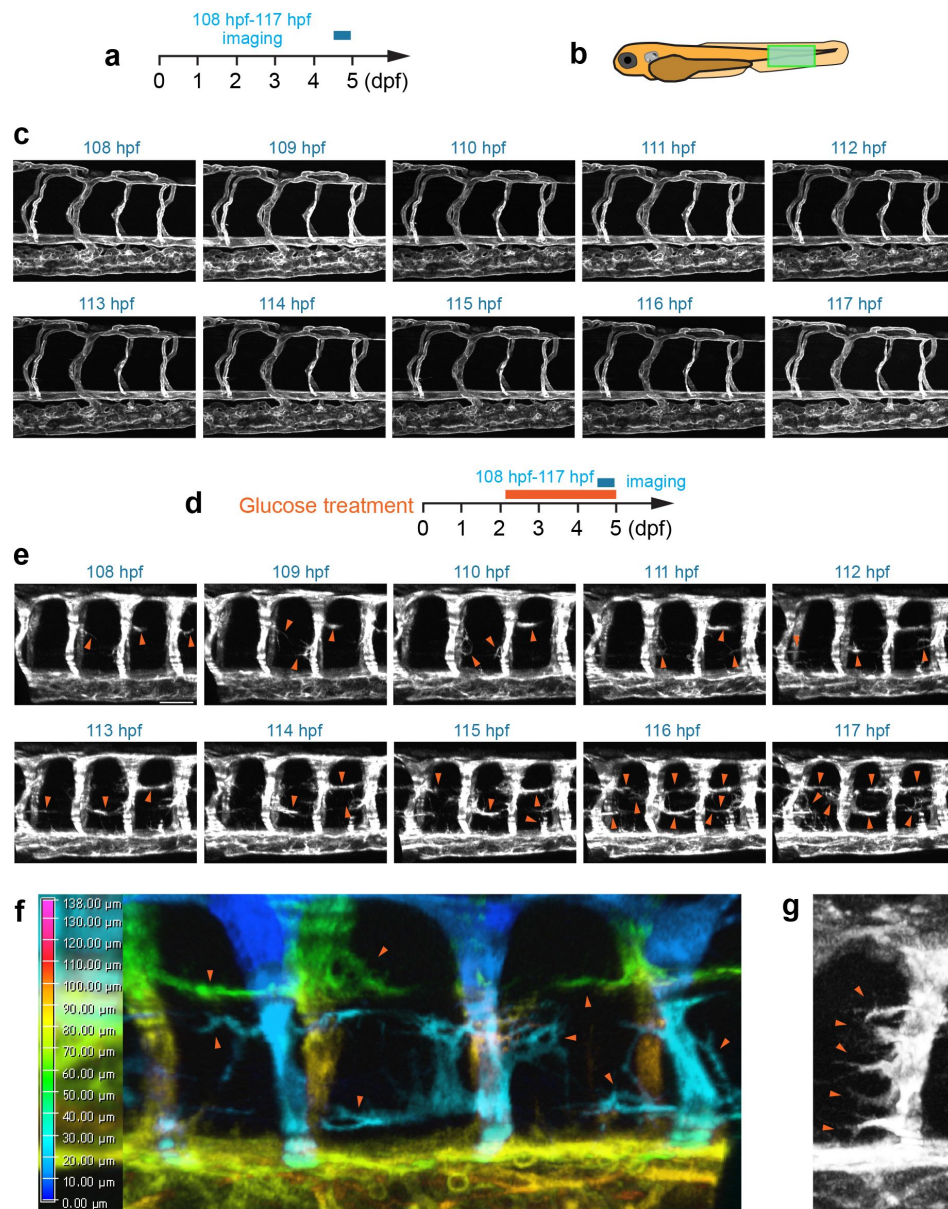


Figure 3

High glucose treatment induced endothelial differentiation into tip cell-like cells.

a, A diagram showing the confocal time-lapse imaging time window. **b**, A diagram indicating the imaging position of the zebrafish embryos. **c**, Confocal time-lapse imaging analysis of blood vessels in control *Tg(fli1aEP:EGFP-CAAX)^{ntu666}* embryos. **d**, A diagram showing the glucose treatment time window and confocal time-lapse imaging time window. **e**, Confocal time-lapse imaging analysis of blood vessels in glucose-treated *Tg(fli1aEP:EGFP-CAAX)^{ntu666}* embryos. Arrowheads indicate the ectopic angiogenic branches. **f**, A snapshot of confocal time-lapse imaging analysis of blood vessels in glucose-treated *Tg(fli1aEP:EGFP-CAAX)^{ntu666}* embryos. Z stacks were used to make 3D color projections, where blue represents the most proximal (closest to the viewer), and red represents the most distal (farthest from the viewer). Arrowheads indicate ectopic angiogenic sprouts. **g**, A snapshot of confocal time-lapse imaging analysis of an ISV in glucose-treated *Tg(fli1aEP:EGFP-CAAX)^{ntu666}* embryos. Arrowheads indicate ectopic angiogenic sprouts.

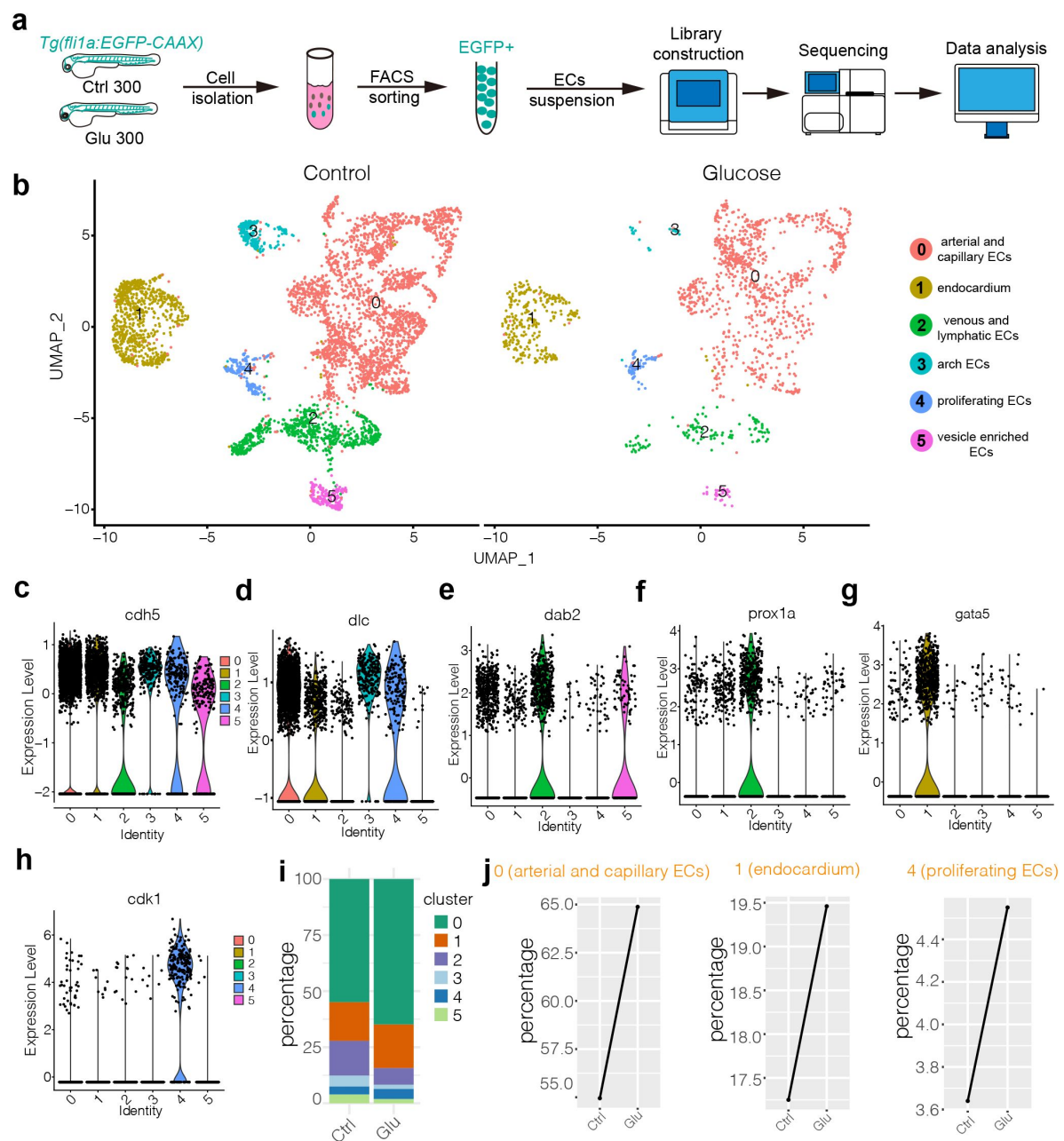


Figure 4
Single-cell transcriptome sequencing analysis of endothelial cells in control and high glucose treated embryos.

a, Schematic diagram of the single-cell sequencing process. 300 embryos in the control group and 300 embryos in the high glucose group were used, and ECs were sorted by GFP fluorescent using FACS technology. **b**, The measured cells were divided into 6 individual clusters based on gene expression profiles using UMAP. **c-h**, The violin plots of some endothelial cell marker genes. **i**, The proportion of ECs in each cluster of the control and high glucose groups. **j**, Changes of ECs percentage in arterial and capillary ECs, endocardium, and proliferating ECs of control and high glucose group.

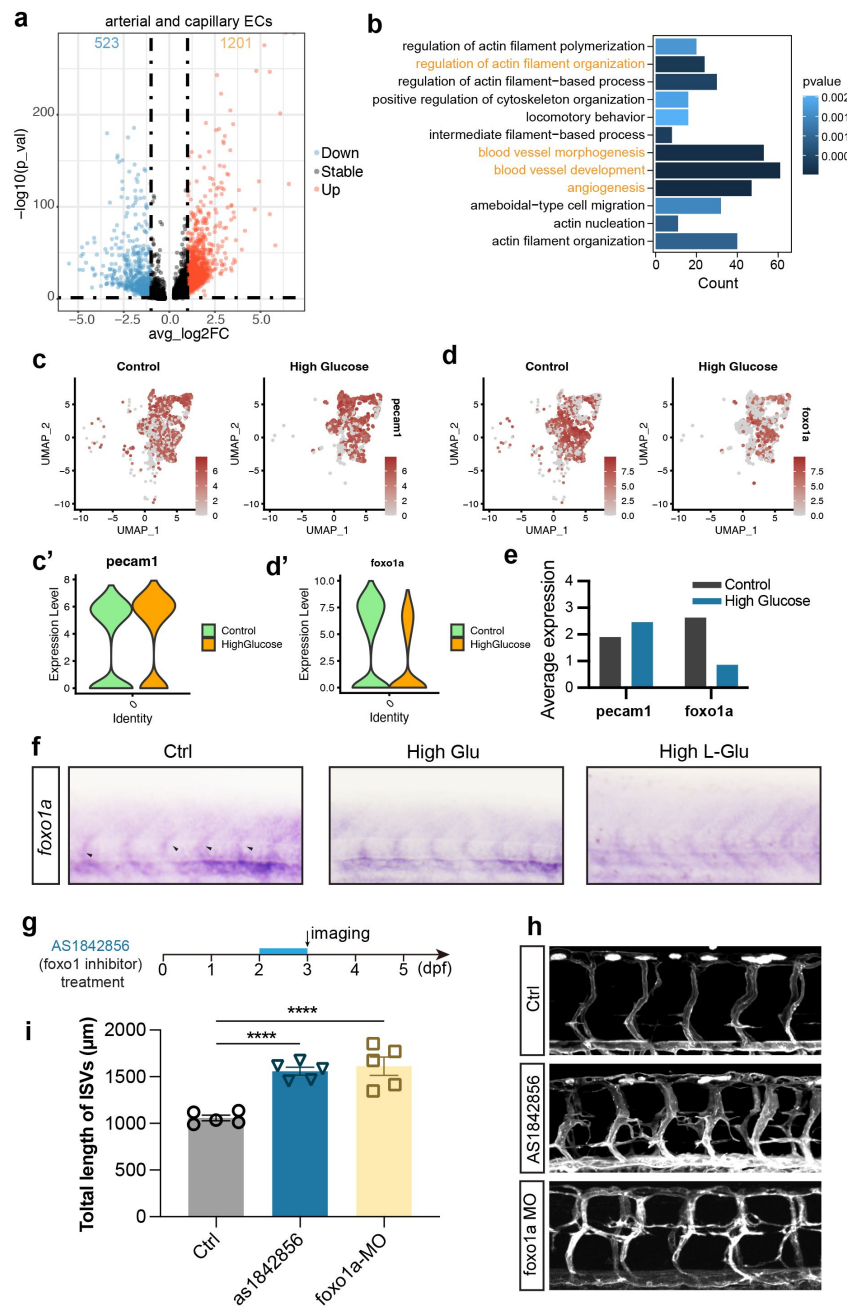


Figure 5

Foxo1a was involved in the excessive angiogenesis induced by high glucose treatment.

a, The volcano plot of differential expression genes in arterial and capillary ECs. The avg_log2FC greater than 1 was considered significant, including 523 down-regulated genes (blue dots) and 1201 up-regulated genes (red dots). **b**, GO analysis of 523 down-regulated genes in arterial and capillary ECs. **c**, The feature plot of ECs marker gene *pecam1* of control and high glucose group in arterial and capillary ECs. **c'**, The violin plot of ECs marker gene *pecam1* of control and high glucose group in arterial and capillary ECs. **d**, The feature plot of gene *foxo1a* of control and high glucose group in arterial and capillary ECs. **d'**, The violin plot of gene *foxo1a* of control and high glucose group in arterial and capillary ECs. **e**, Average expression of gene *pecam1* and *foxo1a* in control and high glucose group. **f**, Whole-mount in situ hybridization analysis of *foxo1a* in control, high glucose, and high L-glucose treated embryos. **g**, A diagram showing the *foxo1* inhibitor treatment time window. **h**, Confocal imaging analysis of control embryos, AS1842856 treated embryos, and *foxo1a* MO-injected embryos. **i**, Statistical analysis of the total length of ISVs in control embryos, AS1842856 treated embryos, and *foxo1a* MO-injected embryos. *t*-test, **** $p < 0.0001$.

treatment with high D-glucose and L-glucose (**Figure 5f**). To verify whether the downregulation of Foxo1a led to excessive angiogenesis in zebrafish embryos, we performed loss-of-function experiments targeting *foxo1a*. AS1842856, a cell-permeable inhibitor that has been reported to block *foxo1* transcription activity [36], was administered to zebrafish embryos at 48 hpf and the imaging was performed at 72 hpf. The results revealed significantly excessive angiogenesis in AS1842856 treated embryos compared with the control group, consistent with the results obtained from *foxo1a* MO injection (**Figure 5g-i**).

Monosaccharides induced excessive angiogenesis through the *foxo1a-marcksl1a* pathway

A previous study has reported that *marcksl1a* overexpression in ECs in zebrafish led to a significant increase in filopodia formation, similar to the phenotype we observed in response to high glucose treatment [37]. Our analysis of the single-cell sequencing data revealed a significant upregulation of *marcksl1a* in arterial and capillary ECs following high glucose treatment, compared to the ECs marker gene *kdrl* (**Figure 6a, b**). The real-time qPCR and ISH experiments further confirmed the elevated expression levels of *marcksl1a* following high D-glucose and L-glucose treatment (**Figure 6c, d**). Then, by constructing the transgenic zebrafish line *hsp70l:marcksl1a-p2A-mCherry::Tg(fli1a:EGFP-CAAX)^{ntu666}*, we conducted the experiments to overexpress *marcksl1a* in zebrafish and subsequently observed the vascular developmental phenotype. After one hour of heat shock for 1 hour at 24 hpf and confocal imaging analysis at 72 hpf, significantly increased blood vessel formation was observed in embryos overexpressing *marcksl1a*, compared with the control group (**Figure 6e-g**).

Given the results obtained from *marcksl1a* overexpression and loss of function of *foxo1a*, we hypothesized that *marcksl1a* might be a target gene of Foxo1a. Therefore, we investigated the impact of Foxo1 inhibition on *marcksl1a* expression in zebrafish embryos. As expected, qPCR analysis revealed that inhibition of Foxo1 by AS1842856 resulted in upregulation of *marcksl1a* expression (**Figure 7a, b**), which suggested that Foxo1a might negatively regulate *marcksl1a* transcription in zebrafish. To further confirm it, we performed the Chromatin Immunoprecipitation (ChIP) experiment to validate the potential binding interaction between Foxo1 and *marcksl1a*. Since the amino acid sequence and DNA binding motifs of Foxo1 are highly conserved between zebrafish and mice (Supplementary Figure 11), we analyzed the 3 kb promoter region of *marcksl1a* to search the binding site sequence of mouse FOXO1 presented in the JASPAR database. Two candidate binding sites (BS) were found at -265 to -275 (BS1) and -153 to -163 (BS2) nucleotides upstream of the TSS of *marcksl1a* (**Figure 7c**) and then used for the ChIP-PCR assay detection. The results showed that Foxo1a was enriched in both the predicted binding sites of *marcksl1a* (**Figure 7d**) in zebrafish.

Additionally, we microinjected *marcksl1a* MO into the 1-cell stage *Tg(fli1a:EGFP-CAAX)^{ntu666}* embryos, which were then treated with high levels of D-glucose and L-glucose. The findings revealed that the knockdown of *Marcksl1a* could effectively mitigate the excessive angiogenesis caused by high D-glucose or high L-glucose treatment, resembling the rescue effect observed with VEGFR inhibitor lenvatinib (**Figure 7e-k**, Supplementary Figure 11). These results suggested that monosaccharides induced excessive angiogenesis through the Foxo1a-*marcksl1a* pathway in zebrafish embryos.

Discussion

In this study, we successfully established a new zebrafish model with significant excessive angiogenesis, resembling the hyperangiogenic characteristics observed in PDR more closely than previously established models [30, 32]. Seung-Hyun Jung et al. have described a short-term zebrafish model for diabetic retinopathy (DR) induced by high glucose, which exhibited blood

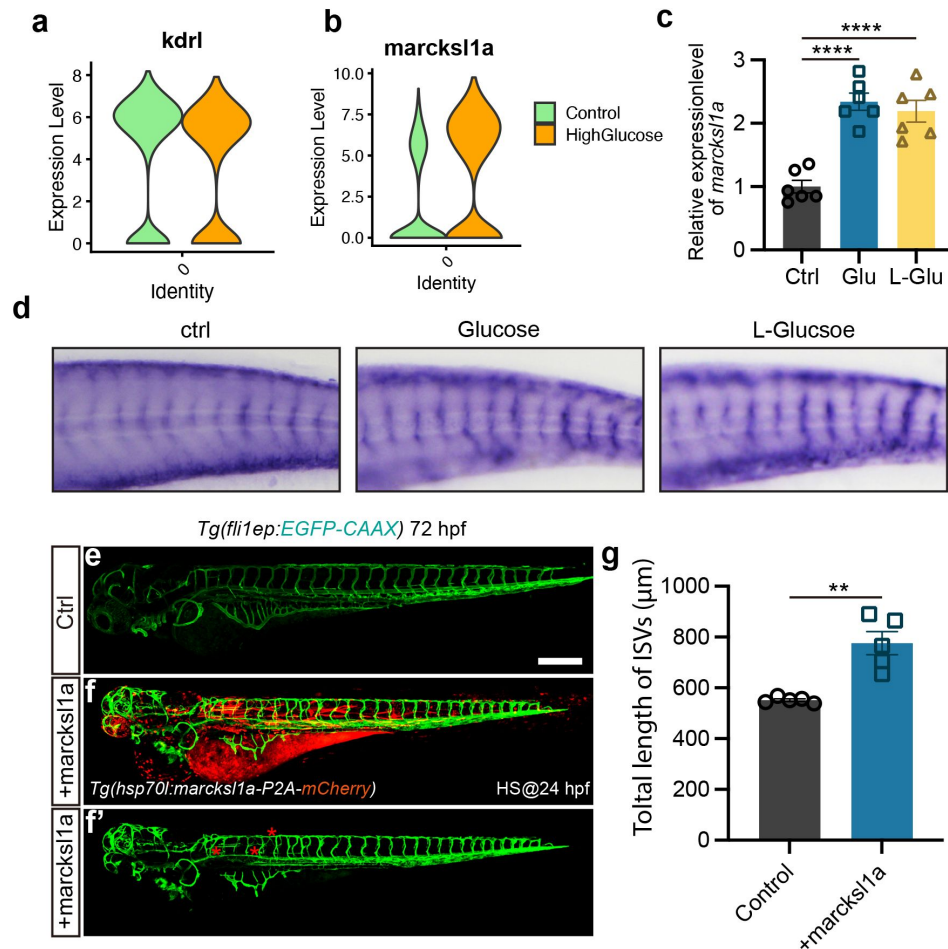


Figure 6

Marcksl1a over-expression induced excessive angiogenesis in zebrafish embryos.

a, The violin plot of ECs marker gene *kdrl* of control and high glucose group in arterial and capillary ECs. **b**, The violin plot of gene *marcksl1a* of control and high glucose group in arterial and capillary ECs. **c**, Real-time PCR analysis of *marcksl1a* expression in control, high glucose, and high L-glucose treated embryos. *t*-test, **** $p < 0.0001$. **d**, Whole-mount in situ hybridization analysis of *marcksl1a* in control, high glucose, and high L-glucose treated embryos. **e-f**, Confocal imaging analysis of blood vessels in control and *hsp70l:marcksl1a-P2A-mCherry* injected *Tg(fli1aEP:EGFP-CAAX)^{ntu666}* embryos. **g**, Statistical analysis of the total length of ISVs in control and *hsp70l:marcksl1a-P2A-mCherry* injected embryos. *t*-test, ** $p < 0.01$.

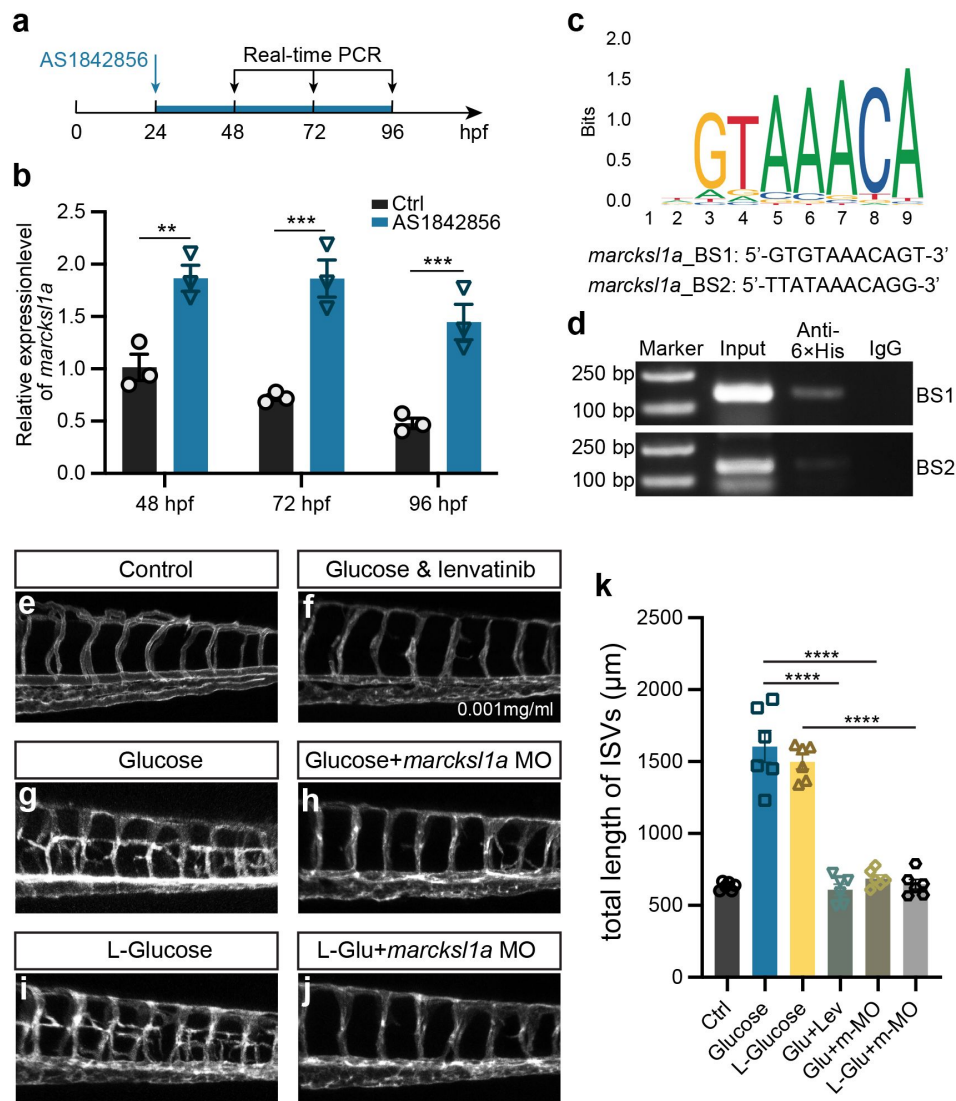


Figure 7

Noncaloric monosaccharides induced excessive angiogenesis through foxo1a-marcksl1a signal in zebrafish embryos.

a, A diagram showing the Foxo1 inhibitor treatment time window. **b**, Real-time PCR analysis of *marcksl1a* expression in control and AS1842856 treated embryos. Two-way ANOVA, $^{**}p<0.01$, $^{***}p<0.001$. **c**, A sequence logo of Foxo1 binding sequence presented in JASPAR database (<https://jaspar.genereg.net/>) and two candidate binding sites at the upstream of transcription start site (TSS) of *marcksl1a* in zebrafish. **d**, Results of the ChIP-PCR assay indicated that BS1 and BS2 are Foxo1a-binding sites of *marcksl1a* in zebrafish. Input sonicated genomic DNA samples without immunoprecipitation as a positive control. IgG, sonicated, and IgG-immunoprecipitated genomic DNA samples as a negative control. **e-j**, Confocal imaging analysis of blood vessels in control, high glucose, high glucose & Lenvatinib, high glucose+*marcksl1a* MO, high L-glucose and high L-glucose+*marcksl1a* MO groups. **k**, Statistical analysis of the total length of ISVs in the groups in figure e-j, respectively. one-way ANOVA, $^{****}p<0.0001$.

vessel defects[30]. However, these defects were limited to the disruption of tight junctions and dilation of hyaloid-retinal vessels [30], without the excessive angiogenesis and vascular blockage observed in PDR and our established model. Additionally, although Kristina Jørgens et al. have observed the hyperbranching of small vessel structures originating from the upper part of ISVs, growing horizontally towards and partially connecting to the neighboring ISVs Field [32], the angiogenic sprouts did not form a more complex structure that was observed in our research.

The excessive development of immature blood vessels represents a significant pathological condition in the progression of DR and nephropathy [38, 39]. Hyperglycemia has been considered one of the most causal factors causing vascular damage, including excessive angiogenesis. However, the exact mechanism through which hyperglycemia impairs the blood vessels is not well determined. To gain more insights into it, we performed the analysis of single-cell transcriptomic sequencing data of the endothelial cells isolated from D-glucose-treated embryos. The findings revealed an increased ratio of tip cells and proliferating ECs, accompanied by the altered expression of various angiogenic genes in the ECs of D-glucose-treated embryos.

Foxo1 has been validated to be essential for sustaining the quiescence of endothelial cells, with involvement in metabolism regulation [34, 40]. Moreover, it also plays important roles in diabetic microvascular complications including DR[41, 42]. Here, by combining the single-cell transcriptomic sequencing data analysis and experimental validation, we identified the transcription factor Foxo1a, which was significantly down-regulated in the embryos treated with high glucose, responsible for the excessive angiogenesis. Additionally, our result further revealed that Foxo1a exerts its regulatory function during this process by down-regulating its target gene *marcksl1a*, regardless of whether the embryos were treated with D-glucose or L-glucose. Taken together, our results suggested that both caloric and noncaloric monosaccharides treatment could lead to excessive angiogenesis by promoting the differentiation of quiescent endothelial cells into tip cells through the *foxo1a-marcksl1a* pathway.

The consumption of ASB has been linked to the occurrence and development of cardiovascular disease in previous studies [12, 16-23]. However, the potential mechanisms underlying the association have not been well documented. In recent years, positive associations between ASB and cardiovascular disease have been proposed, possibly due to several plausible factors, including the potential impact of ASBs on central nervous system circuits, gut hormone secretion, and gut microbiota [43-45]. Additionally, it has been hypothesized that the ASBs might stimulate appetite and increase calorie intake [43, 44].

For a long time, there has been considerable debate and conflicting opinions regarding how specific sugars affect the development of type 2 diabetes rather than excess calories per se [46, 47]. In this study, we have provided new evidence indicating that the administration of noncaloric monosaccharides leads to significant excessive angiogenesis, suggesting that the excessive angiogenesis may not be only attributed to the caloric properties. Since excessive angiogenesis is the major pathological feature of diabetic retinopathy and nephropathy, our findings are in support of a possible biological mechanism underlying the positive associations between noncaloric monosaccharides and microvascular complications associated with type 2 diabetes, suggesting that the noncaloric monosaccharides might not be suitable for ASB consumption.

Surprisingly, no notable abnormalities were observed in the vessels of embryos treated with disaccharides, including lactose, maltose, and sucrose, which is consistent with the previous study stating that intakes of sucrose, lactose, and maltose were not significantly associated with the risk of type 2 diabetes [48]. This finding implied that the effects induced by monosaccharides cannot be attributed to the osmotic pressure of the surrounding medium. Furthermore, despite the

potential conversion of these disaccharides into monosaccharides, the restricted reaction rate may maintain them within a safe concentration range that is not harmful to the vessels in a short period.

In conclusion, to investigate the effects of monosaccharides on vascular development, we established a zebrafish model by treating the embryos with high concentrations of monosaccharides. Based on this model, we observed significant excessive angiogenesis induced by glucose and noncaloric monosaccharides, initiated by activating the quiescent endothelial cells into proliferating tip cells. The effects of monosaccharides on inducing excessive angiogenesis were then proved to be mediated by the *foxo1a-marcksl1a* pathway. The results have provided novel insights into the roles of noncaloric monosaccharides in human health and the underlying mechanisms.

Materials and methods

Zebrafish

Care and breeding of zebrafish were carried out as previously described [49]. Animal experiments were conducted according to local institutional laws and Chinese law for the Protection of Animals. The following transgenic strains were used: *Tg(fli1aEP:EGFP-CAAX)^{ntu666}* and *Tg(kdrl:ras-mCherry)* [33]. Embryos were obtained through natural mating and maintained at 28.5°C. The stages of zebrafish embryos are defined as previously described [49]. Embryos were treated with 0.2 mM 1-phenyl-2-thiourea (PTU, Sigma, P7629) to block pigmentation for further imaging analysis.

Monosaccharides and drug treatment

The D-glucose (Sigma, G7021-100g), L-glucose (Sigma, G5500-1g; J&K, 981195-1g), D-Fructose (Sigma, F0127-100g), L-Rhamnose monohydrate (Aladdin, R108982), D-Sorbitol (Sigma, S1876-100g), D-Mannitol (Sigma, M4125-100g), D-(-)-Ribose (Sigma, V900389-25g), L-(+)-Arabinos (Sigma, V900920-25g), Mannose (Sigma, M2069-25g) and sucrose (Sigma, V900116-500G) were dissolved in E3 solution. Zebrafish embryos at 24 hpf to 48 hpf were placed in 24-well plates (ten embryos per well) and immersed in the solution at the presetting concentrations and time windows. Then, put it in a 28 °C incubator for cultivation. Five days before embryonic development, a stereo fluorescence microscope and a laser confocal microscope were used to observe the changes in blood vessel phenotype. For the drug treatment, the embryos were co-incubated in glucose with lenvatinib (Selleck, S1164-5MG) from 48 hpf to 96 hpf. Foxo1 inhibitor AS1842856 (MCE, HY-100596) was dissolved in DMSO and stored at -80°C and diluted with E3 solution when used. The same concentration of DMSO was used as a negative control.

Glucose concentration measurement

Glucose concentration in the embryo was measured as described previously [50]. Embryos that developed to 75% epiboly were selected and transferred to 24-well plates (ten embryos per well) and immersed in the solution at the presetting concentrations and time windows. For glucose concentration measurement, embryos (n=20) were transferred to a new 1.5 mL tube, rinsed three times with 1×PBS, and immersed in ice for the following experiments. Discard the PBS as much as possible, embryos homogenized using a hand homogenizer, and centrifuged at 14,000×g for 10 min. 1.5 µL of the supernatant was used to measure the total free-glucose level using a glucometer (Baye, 7600P).

Whole-mount in situ hybridization (WISH)

Whole-mount in situ hybridization and the preparation of antisense RNA probes were performed as described in the previous protocol [51]. Briefly, the *marcksl1a* and *foxo1a* cDNA fragments were cloned with the specific primers listed below using the wild-type embryo (AB) cDNA library. Probes were synthesized using the in vitro DIG-RNA labeling transcription Kit (Roche, 11175025910) with linearized pGEM-T easy vector containing *marcksl1a* or *foxo1a* gene fragment as the templates. Synthesized probes were purified with LiCl (Invitrogen, AM9480) and diluted to 1 ng/μL for hybridization. Zebrafish embryos were collected and fixed with 4% paraformaldehyde (PFA) overnight at 4°C, then dehydrated with methanol gradients and stored at -20°C in 100% methanol. The hybridization result was detected with anti-DIG-AP antibody (1:2000, Roche, 11093274910) and NBT/BCIP (1:500, Roche, 11681451001). After hybridization, images of the embryos were captured with an Olympus stereomicroscope MVX10. The primers are listed below:

marcksl1a-probe-forward:5'- AGG ATG GGT GCT CAG TTG AC-3'

marcksl1a-probe-reverse:5'- GCT GGC GTC TCA TTG GTT TC-3'

foxo1a-probe-forward:5'-GCA ACA CAG GAT TTC CCC AC-3'

foxo1a-probe-reverse:5'-CAC AGG TGG CAC TGG AAG G-3'

Single-cell gene expression profile analysis

Cell Ranger 3.0.2 (<https://github.com/10XGenomics/cellranger>) was used to convert the raw sequencing data to a single-cell level gene count matrix. The clustering of single cells and the marker genes in each cluster were analyzed by Seurat 3.0 (<https://satijalab.org/seurat/install.html>) [52]. Several criteria were applied to select the single cells, including only keeping the genes that are expressed (Unique Molecular Identifiers or UMI larger than 0) at least in 3 single cells, selecting single cells with the number of expressed genes at the range between 500 and 3000, and requiring the percentage of sequencing reads on mitochondrial genome being less than 5 percentage. Furthermore, SCTransform method [53] was applied to remove technical variation, and ClusterProfiler [54] was used to do the Gene Ontology enrichment analysis based on the marker genes of each cell cluster. Detailed information about the data processing can be found in the source code of this project (<https://github.com/gangcai/ZebEndoimmune>).

Gene expression analysis by quantitative real-time PCR

Total RNA was extracted from zebrafish embryos using TRIzol™ (Invitrogen, 15596026), and stored at -80°C. The cDNA was then synthesized using the HiScript II Q RT SuperMix for qPCR Kit (Vazyme, R223-01) according to the manufacturer's instructions. Quantitative PCR was performed in triplicates using the Taq Pro Universal SYBR qPCR Master Mix (Vazyme, Q712-02) on a real-time PCR detection system (StepOne™ Real-Time PCR Systems). The primers used for Real-time PCR analysis are as follows:

ef1a-Qpcr-F:5'- CTT CAA CGC TCA GGT CAT CA -3'

ef1a-Qpcr-R:5'- CGG TCG ATC TTC TCC TTG AG -3'

marcksl1a-Qpcr-F:5'- CCG TGG CTG ATA AAG CCA AT -3'

marcksl1a-Qpcr-R:5'- CTC CCT CCT CCG TTT TTG GG -3'

Transgenic and heat shock

The *Tg(fli1aEP:EGFP-CAAX)^{ntu666}* line was established using a construct *fli1aEP:EGFP-CAAX*, which was generated using multisite Gateway technology, the tol2 kit as previously described [55]. The 5' Entry p5Efl1ep (#31160) purchased from Addgene was originally from Nathan Lawson Lab [56]. Three entry clones and the pDestTol2pA2 destination vector were used to generate the expression construct by LR recombination reaction as described in the Multisite Gateway Manual book. The expression constructs were synthesized by GENEWIZ company. The zebrafish embryos were immersed in a 37 °C water bath for 1 hour for heat shock. Around 75 pg of expression plasmid DNA and 25 pg tol2 transposase mRNA were premixed and microinjected into single-cell fertilized eggs.

Chip-PCR

Embryos injected with *hsp70l:foxo1a-6×His-P2A-mCherry* were collected at 72 hpf after heat shock treatment. According to the manufacturer's instructions, the ChIP-PCR assay was performed using the Chromatin Immunoprecipitation (ChIP) Assay Kit (Millipore, 3753379). The genomic DNA crossed with Foxo1a protein was immunoprecipitated by using 5 µg Anti-6×His tag antibody (abcam, ab213204). Antibody against IgG was used as a negative control. The semiquantitative PCR was performed with KODfx (TOYOBO, KFX-101) at the following conditions: 94°C for 5 min; 35 cycles of 98°C for 10 s, 55°C for 30 s, 68°C for 10 s; 68°C for 10 min. The PCR primers used for the predicted binding sites (BS) are as follows:

Marcksl1a-BS1-forward:5'- CCC TTT TTC AAA AGT GAG TTT GAG -3'

Marcksl1a -BS1-reverse:5'- GGA GCT TCA TCT GCC CCA TT -3'

Marcksl1a -BS2-forward:5'- CGG TTT CCA GCT TTC TTC AGA A -3'

Marcksl1a -BS2-reverse:5'- TCT CAA ACT CAC TTT TGA AAA AGG G -3'

Imaging analysis

For confocal imaging of blood vessels in fluorescence protein labeled transgenic zebrafish embryos, they were anesthetized with egg water/0.16 mg/mL MS222 (Sigma, A5040)/1% PTU and embedded in 0.5-0.8% low melting agarose. Confocal imaging was performed with a Nikon A1R HD25 Confocal Microscope. Analysis was performed using Nikon-NIS-Elements software. The bright field images were acquired with an Olympus DP71 camera on an Olympus stereomicroscope MVX10.

Statistical analysis

Statistical analysis was performed with a student's t-test. All data is presented as Mean ± SEM; p < 0.05 was considered statistically significant.

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Conflicts of interest/Competing interests

The authors declare that they have no conflicts of interest

Availability of data and material (data transparency)

All the experimental materials generated in this study are available from the corresponding authors upon reasonable request.

Authors' contributions

Dong Liu, Xuchu Duan, Xia Liu, and Gangcai Xie conceived and designed the experiments and wrote the manuscript. Xiaoning wang, Jinxiang Zhao, Jiehuan Xu, Bowen Li, and Gangcai Xie performed the experiments and analyzed the data. All authors read and approved the final manuscript.

Ethics approval

All zebrafish experimentation was carried out following the NIH Guidelines for the care and use of laboratory animals (<http://oacu.od.nih.gov/regs/index.htm> ) and ethically approved by the Administration Committee of Experimental Animals, Jiangsu Province, China (Approval ID: 20180905-002).

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Not applicable

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Editors

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Reviewer #1 (Public Review):

Summary:

This manuscript successfully established a new short-term model of diabetic retinopathy by treating zebrafish embryos with high concentrations of monosaccharides, resembling the hyperangiogenic characteristics observed in proliferative diabetic retinopathy in patients. They found that excessive angiogenesis induced by glucose and noncaloric monosaccharides was achieved by activating the quiescent endothelial cells into proliferating tip cells. Importantly, the authors further confirmed the effects of monosaccharides on inducing excessive angiogenesis were mediated by the foxo1a-marcksl1a pathway.

Strengths:

These results showed the potentially detrimental effects of the noncaloric monosaccharides on blood vessel function and provided novel insights into the underlying mechanisms.

Weaknesses:

The mechanism of noncaloric monosaccharides inducing excessive sprouting angiogenesis is not solid enough.

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Reviewer #2 (Public Review):

In the manuscript entitled "Noncaloric monosaccharides induce excessive sprouting angiogenesis in zebrafish via foxo1a-marcksl1a signal". Liu et al. observed that glucose and noncaloric monosaccharides can prompt an excessive formation of blood vessels, particularly

intersegmental vessels (ISVs). They propose that these branched vessels arise from the ectopic activation of quiescent endothelial cells (ECs) into tip cells. Moreover, through single-cell transcriptome sequencing analysis of embryonic endothelial cells exposed to glucose, they noted an increased proportion of arterial and capillary endothelial cells, proliferative endothelial cells, along with a series of upregulated genes in categories of blood vessel morphogenesis, development, and pro-angiogenesis. The authors provide evidence suggesting that caloric and noncaloric monosaccharides (NMS) induce excessive angiogenesis via the foxo1a-Marcksl1a pathway.

The authors address an important problem about the effects of artificially sweetened beverages such as noncaloric monosaccharides on blood vessels. However, the study lacks adequate experimental data and comprehensive analyses to support the mechanistic conclusions, which require extensive revisions.

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Reviewer #3 (Public Review):

The authors have investigated the effect of noncaloric monosaccharides on angiogenesis in the zebrafish embryo. These compounds are used as substitutes of sugars to sweeten beverages and they are commonly used by diabetic patients. The authors show that noncaloric monosaccharides and glucose similarly induce excessive blood vessel formation due to the increased formation of tip cells by endothelial cells. The authors show that this excessive angiogenesis involved the foxo1a-marcksl1a pathway.

A limitation of the study is that the mechanism of angiogenesis in the retinal circulation and in peripheral vasculature is certainly different.

This result suggests that these noncaloric monosaccharides share common side effects with glucose. Consequently, more caution should be taken with regard to the use of these artificial sweeteners. This work is of interest for better management of diabetes.

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